Adaptive Reliability Reserves: Learning Risk-Adjusted Backup for Residential Battery Fleets

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Abstract

Residential battery virtual power plants promise each household outage backup while monetizing the same batteries in wholesale markets. Industry practice resolves this tension with a fixed state-of-charge floor (typically 20%) plus a binary weather-alert "storm mode." We ask whether the backup reserve should instead be a risk-adjusted sequential decision. We formalize the customer promise as a per-home reliability-risk budget—expected unserved backup energy per year—and compare fixed floors, alert rules, quantile reserves, risk-priced model-predictive control, and a learned policy that sets fleet-level dispatch and reserve targets above a feasibility-preserving disaggregation layer. Evaluation uses a public replica of a Texas home-battery fleet: NREL ResStock homes, ERCOT prices (2019–2026), and ORNL EAGLE-I county outage records (2014–2025). [TODO: headline numbers: frontier improvement at equal realized reliability; behavior under Winter Storm Uri; fleet-size/composition shift.] Price spikes and mass outages are nearly disjoint in ERCOT (2021–2025: 107 spike days, 28 mass-outage days, 4 joint—all Winter Storm Uri), so hazard-conditioned reserves are close to free most of the year; we characterize where they fail.

1 Introduction

[TODO: Tension: dual-use storage. Base-style fixed 20% floor + storm hold; Tesla Storm Watch; Sunrun 20%. Why always 20%? Contributions: (1) reserve as learned allocation of a per-home reliability budget, nesting industry rules and quantile reserves; (2) HomeFleet-ERCOT, a public benchmark coupling real prices, real outage records, and realistic homes; (3) evaluation under market, outage, and fleet shift with failure analysis.]

2 Problem: backup promises as a risk budget

[TODO: Firm margin (A&Z); battery dynamics; outage process from EAGLE-I via quantile matching; unserved energy; per-home budget CMDP; multiplier = implied VoLL; special cases: fixed floor, storm mode, quantile tier, chance-constrained MPC.]

3 Controllers

[TODO: DP/MPC family with shared forecasts; risk-priced MPC (same objective as RL); PPO + disaggregation layer; action = aggregate power + reserve multiplier; feasibility by construction.]

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4 Benchmark and protocol

[TODO: Data sources and splits (train $\leq$2023, val 2024, test 2025–26, event holdouts); the 2×2 spike-outage event design; metrics: margin/home-week, capture vs oracle, CVaR_5, ride-through and 5-hour promise rates, unserved kWh; common random numbers.]

5 Results

[TODO: Fig 2 orthogonality; Fig 3 margin-reliability frontier; Fig 4 event panels (Uri, Beryl, Aug-2023 heat); Fig 5 shift (years, RTC+B, fleet size/composition, dropout); Table tails + failure taxonomy. Early findings to confirm: risk-priced MPC beats fixed floor in every spike window at equal reliability; storm-watch loses money in heat waves but is the best hedge in Uri; forecast-based controllers all lose money in Uri (REP exposure).]

6 Limitations and deployment notes

[TODO: County-level hazard; price-taker; AMY2018 loads not cold-amplified; ADER caps; sim-to-real (6.7% precedent); what an operator would need before eyes-off.]

LLM usage disclosure

[TODO: Statement per NeurIPS 2026 policy: LLM-assisted coding and editing; all citations and numbers verified by the authors.]

References

A Domain checklist

1. **Physics feasibility.** [TODO: SOC/inverter/backup feasibility by construction (disaggregation layer); AC power-flow verification on an NREL SMART-DS Austin feeder; violation statistics.]
2. **Out-of-distribution evaluation.** [TODO: held-out years incl. post-RTC+B prices; event holdouts excluded from training; unseen homes; fleet size/composition shift.]
3. **Failure modes.** [TODO: Uri panel; tail stats (CVaR_5, worst week); failure taxonomy by event class.]
4. **System-level effect.** [TODO: hazard-model quality $\rightarrow$ downstream margin/reliability (value-of-information bracket: alerts-only vs oracle hazard).]
5. **Data and code.** [TODO: all-public data; anonymized repo link; full release on acceptance.]

B Extended results

[TODO: per-window tables; degradation-model sensitivity; ablations.]